Dr Seamus Murphy
TÜV SÜD, 2187 Comox Ave,
Comox, British Columbia V9M 1P5, Canada
seamusrobertmurphy@gmail.com
ORCID 0000-0002-1792-0351

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Dr Christopher Ranger
Editor-in-Chief
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Royal Entomological Society

RE: Resubmission of AFE(2026)5804 as a new manuscript

Dear Dr Ranger,

Thank you for the decision of 22 September 2026 on manuscript AFE(2026)5804, “Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*)”, and for the invitation to resubmit it as a new manuscript. You asked that the Introduction and Discussion be revised to include the existing literature and that the novelty of the study be stated explicitly, and Dr MacQuarrie asked that the Discussion link the findings to the biology, ecology or management of the beetle rather than restate the Results. The attached response quotes each comment and gives the section of the revised manuscript where it is addressed.

The literature was gathered in two searches under written protocols, and the reference list grew from 30 to 61 works. The Introduction now opens with the beetle as a native disturbance agent of pine forests and the history of its outbreaks, and it builds each of the three questions from studies of how the beetle lands on, attacks and kills a tree, survives the winter, flies and spreads. The Discussion sets each result against that literature and closes on what it means for managing stands.

The wind terms were rebuilt. The station wind of the submitted manuscript, one value for the whole site in each period, was joined by a terrain wind model run over the site, the surrounding valley and the whole Purcell Trench from the reanalysis, checked hour by hour against four valley stations, corrected for the constant share of speed the model removes, and used with its direction limit stated. With that wind, and crown closure as the measure of openness, wind lowered attack where the canopy was open and raised it where it was closed, which is the plume disruption the second question asked about, and the revised manuscript reports it as such.

Two defects in the submitted data were also repaired. The attack maps were rebuilt with a classifier validated on field plots held out whole, and the terrain-adjusted wind field, which had covered less than a third of the study area, now covers all of it. The design is unchanged, with attack mapped once a year for the refugia questions and every sixteen days for the wind test. The spatial pattern of attack is now described with join counts and a Moran’s I correlogram before any model is fitted, the models follow the order of Aukema et al. (2008) in choosing the terms for attack in and around each cell the year before ahead of any environmental term, and the results are checked under a latent spatial field and at three coarser grains. In the revised analysis the refugia on this landscape were thin stands with little host and ground shaded by terrain, wind lowered attack only in open stands, and ground facing the afternoon wind had the least attack, an effect that depended on the grain and place at which it was read. The Results answer the three questions in turn and the Discussion sets each answer against the literature in the same order. The response states which findings held and which changed.

The novelty of the study is what it did. It tested the three refugia mechanisms of Krawchuk et al. (2020) together on one landscape, beside explicit terms for attack nearby and in the previous year, and it tested the two ways wind could act, breaking up the pheromone plume, with a terrain-adjusted wind field over the afternoon flight hours of each sixteen-day period, and depositing beetles on sheltered ground, with a wind shelter index computed from the terrain.

The study grew out of a companion study of forest regeneration on the same ground, Murphy, Leslie, Wilson and Banks (2026), Forest Ecology and Management 618, 123985, whose field plots of beetle-killed pine trained the classifier here and are used with the agreement of its co-authors, as the title page states. The work is original, is not published or under consideration elsewhere, and received no external funding. I am the sole author, I declare no conflict of interest, there is no dispute over the ownership of the data presented, and every contribution has been attributed by authorship or acknowledgement. The derived data and the complete analysis code are public on GitHub.

Yours sincerely,

Dr Seamus Murphy